DPP DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CENTRAL VENOUS PRESSURE MONITORING
DEPARTMENT:	INTENSIVE CARE UNIT
APPLIES TO:	ALL MEDICAL AND NURSING STAFF

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1. PURPOSE

1.1. To provide procedural guidelines for the monitoring of central venous pressure.

1.2. To obtain a basis for fluid replacement in seriously ill patient.

1.3. To estimate blood volume deficits.

1.4. To determine pressure in the right atrium and central veins.

1.5. To evaluate circulatory failure.

2. DEFINITION & ABBREVIATION

2.1. **Central Venous Pressure Monitoring** – a continuous or manual measurement of pressure within the right atrium or in the great veins of the heart.

2.2. **Water Manometer Technique** – a technique used to connect the manometer to an intravenous drip set on one side and via extension tubing filled with intravenous fluid to the patient on the other.

2.3. **Pressure Transducer** – use to obtain a precise and continuous measurement of physiological pressures.

2.4. **Phlebostatic Axis** – it is on the 4th intercostal space in the mid-axillary line on a supine patient.

2.5. **Carpenter's Level** – a mark on patient's chest for future reference.

2.6. **ICU (Intensive Care Unit)** – is a specialized hospital ward that provides constant monitoring and intensive treatment for patients with life-threatening illnesses or injuries, such as severe accidents, heart attacks, strokes, serious infections, or major surgeries.

2.7. **CVP (Central Venous Pressure)** – is a large vein near the heart (vena cavae) and reflects the volume of blood returning to the heart and the heart's ability to pump it out.

2.8. **cm H2O** – Centimeters of Water

2.9. **mmHg** – Millimeter of Mercury

3. RESPONSIBILITY

3.1. **ICU Physician** – write the order for monitoring the CVP.

3.2. **ICU Staff Nurse** – must monitor the CVP and document the procedure.

3.3. **ICU Head Nurse** – ensures that all staff is aware of this policy and are competency tested.

4. POLICY

4.1. Reading of CVP must be performed by competent and qualified Physician and Staff Nurses.

4.2. The equipment needed for CVP measurement must be available at all times.

4.3. Standard Precaution must be observed.

4.4. ICU nurses must be competent in identifying normal and abnormal reading and notify the physician once alteration is noted.

4.5. Each and every reading should be taken with patient's same position.

4.6. Monitor the CVP every 2 hours or as directed by the physician.

4.7. Ensure the transducer level is at the patient's Phlebostatic axis.

5. PROCEDURE

5.1. Physician writes his order for CVP monitoring.

5.2. Identify the patient using two identifiers. (Refer to APP-QPS-0016 Patient Identification).

5.3. Provide Privacy.

5.4. Explain the procedure to the patient.

5.5. Gather all equipment needed.

5.6. Perform Hand Hygiene. (Refer to APPH-IPC-0005 Guidelines in Hand Hygiene).

5.7. Place the patient in supine position without pillow.

5.7. 1. If contraindicated, lower preferably less than 30 degrees. 5.8. Start measuring CVP.

5.8. Temporarily set a Positive End-Expiratory Pressure (PEEP) to zero on a mechanical ventilator to obtain a more accurate reading of the patient's central venous pressure (CVP) if patient's condition allowed.

5.9. Start measuring CVP.

5.9.1. **Water Manometer Method**

5.9.1.1. Connect the rye tubing to the manometer.

5.9.1.2. For manometer set up, secure the water manometer to the IV pole on Zero level of the Phlebostatic axis of the patient and use carpenter's level.

5.9.1.3. Tum on the water manometer stopcock and prime with 0.9% Sodium Chloride.

5.9.1.4. Tum stopcock off to patient and open the manometer.

5.9.1.5. Fill the manometer greater than suspected CVP and ensure that there's no bubble in the manometer tubing.

5.9.1.6. Bubble can alter CVP reading.

5.9.1.7. Open the IV flow between patient and manometer.

5.9.1.8. Start the CVP reading by observing fluid level.

5.9.1.8.1. CVP pressure equals the stabilized level at end-expiration.

5.9.1.8.2. Read on ball midline.

5.9.1.8.3. Fluctuation is normal during inspiration.

5.9.1.9. Turn stopcock off to manometer and resume IV flow of the patient if available.

5.9.2. Hemodynamic Monitoring System

5.9.2.1. Validate the waveform as CVP/RAP on the cardiac monitor.

5.9.2.2. Push the IV spike to 0.9% Sodium Chloride.

5.9.2.3. Place the 0.9% Sodium Chloride in pressure cuff and hang on IV stand.

5.9.2.4. Open the roller clamp and fill IV set by gravity. Tap IV set to free trapped bubbles.

5.9.2.5. Close the roller clamp and connect filled IV set to the monitoring system

5.9.2.6. Pressurize the infusion bag to 300mmHg.

5.9.2.7. Connect the transducer to the IV tubing and fill the cassette system.

5.9.2.8. Tap gently to remove remaining air bubbles.

5.9.2.9. Replace all vented caps and tighten it on side ports of stopcock.

5.9.2.10. Secure transducer on holder or directly on patient's arm with transducer Zeroing port Phlebostatic axis.

5.9.2.11. Connect monitoring system to the patient.

5.9.2.12. Affix appropriate label to the transducer when using multiple transducer.

5.9.2.13. Zero-balance the monitoring system to atmospheric pressure and Calibrate transducer according to monitor manufacturer's instruction.

5.9.2.14. Tum zeroing stopcock handles clockwise and check the quality of the wave form.

5.9.2.15. Allow approximately I minute for the system to ensure that flush Device is operating properly.

5.9.2.16. Replace the cap, tum stopcock closed to the side port and continue the monitoring of CVP wave form.

5.10. If there is abnormal reading:

5.10.1. **Recheck** the procedure as leak may be present and obtain another Measurement.

5.10.2. Notify the physician and manage according to the order.

5.11. Place the patient in comfortable position.

5.12. **Reset the Positive End-Expiratory Pressure (PEEP) on the mechanical ventilator if adjusted to zero for CVP reading.**

5.13. Perform Hand Hygiene. (Refer to APPH-IPC-0005 Guidelines in Hand Hygiene).

5.14. Document the measurement in Nurses Observation Form, Nurses Progress Notes Form and Physician Progress Notes Form in the MCC.

6. MEASUREMENT

6.1. 100% of the medical and nursing staff are well oriented and adheres to this policy.

7. REFERENCES /CROSS- REFERENCES

7.1. http://nursingscrib.com/nursing-notes-reviewer/fundamentals-of-nursing/how-to-measure-cvp-central-venous-pressure/

7.2. https://en.wikipedia.org/wiki/Central_venous_pressure

7.3. Weigand DL 6th edition (2010): AACN procedure manual for critical care, Philadelphia

7.4. MOH-Manual of Nursing Policies and Procedures 2nd edition 2011

7.5. Argon Medical Devices Manufacturers Manual

7.6. https://www.respiratoryupdate.com/members/login.cfm?hpage=Measurement_of_CVP_via_Catheter_Water_Manometer_Technique.cfm

7.7. APP-QPS-0016 Patient Identification

7.8. APPH-IPC-0005 Guidelines in Hand Hygiene

8. ATTACHMENT

8.1. Central Venous Pressure Monitoring Competency Check Off

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design

9.2. Modification to definition and abbreviation.